Thawin Serivivatanavongse

Atlanta, GA | tseriviv3@gatech.edu | linkedin.com/in/thawin

Education

Georgia Institute of Technology Atlanta, GA
Bachelor of Science in Electrical Engineering, GPA: 4.00 *December 2027*
Concentration: Signal & Information Processing | Circuit Technology
Relevant Coursework: Signals & Systems, Analog Electronics, Microelectronic Circuits, Electromagnetics

Experience

Analog/Mixed-Signal IC Design Intern May 2026 – August 2026
Analog Devices *Chonburi, Thailand*

- Implemented a two-stage dynamic comparator for a 5 MS/s 10-bit SAR ADC in TSMC 65 nm and characterized it with Monte Carlo and transient-noise simulation in Cadence ADE Assembler, achieving an offset standard deviation σ_{OS} of 2.21 mV and input capacitance C_{in} of 60 fF.
- Identified the trade-off between input-pair size, offset mismatch, and input capacitance, and traced comparator kickback to a floating tail node during reset; proposed an active tail-reset device to eliminate it.
- Designed asynchronous self-timed logic that generates the comparator clock and SAR bit-cycling signals internally, removing the need for an external high-speed clock.
- Designed and hand-laid-out the CDAC capacitor array in Virtuoso using delta-length MOM capacitors with a 125 aF unit capacitance, sized to balance kT/C noise against chip area.

Undergraduate Research Assistant, RF Beamforming (Rotman Lens) August 2025 – May 2026
Georgia Institute of Technology *Atlanta, GA*

- Developed a MATLAB-based numerical synthesis pipeline for a non-focal 24 GHz Rotman lens, solving nonlinear path-length equations to compute beam, array, and dummy port geometry.
- Generalized the classical tri-focal Rotman lens design to an elliptical contour with multiple pseudo-focal points, reducing peak phase error and improving beam resolution uniformity across a $\pm 30^\circ$ scan range.
- Formulated a constrained nonlinear optimization (gradient-based) that minimizes RMS phase error across 11 pseudo-focal points, converging to a 10^{-9} tolerance under KKT optimality conditions.

Design Verification Engineer January 2024 – Present
Georgia Tech Silicon Jackets *Atlanta, GA*

- Designed a GCD module as a 4-state FSM in SystemVerilog and verified it with a constrained-random testbench and self-checking scoreboards, achieving over 99% code coverage.
- Authored a test plan and built a UVM testbench for a fast divider unit, verifying functional correctness across edge and corner cases with over 99% code coverage.

Projects

Current-Starved Ring Oscillator VCO | *SKY130, Cadence Virtuoso, Spectre* Fall 2026

- Designed a 3-stage current-starved ring oscillator VCO in SKY130 (1.8 V), sizing devices from hand analysis of stage delay, per-stage load, and body-effect threshold, and added a 4-stage tapered buffer to drive a 1 pF load.
- Achieved 501.3 MHz at 174.9 μW (500 μW budget), 48.2% duty cycle, and 114/162 ps rise/fall times at TT, 27 °C, meeting all specs; traced an initial ∼43% duty-cycle error to the ring stages, not the output buffer.
- Verified frequency stays within the 450–550 MHz spec across 0–70 °C (487–519 MHz) and characterized the SS@0 °C to FF@70 °C corner spread (376.5–635.6 MHz) in Cadence Spectre.

Technical Skills

Tools: Cadence Virtuoso, ADE Assembler/Explorer, Spectre, LTspice, Calibre (DRC/LVS), TSMC 65 nm and SkyWater SKY130 PDKs, GitHub, Linux, LaTeX
Lab: Oscilloscope, function generator, DMM, bench power supply, spectrum analyzer, VNA, logic analyzer, soldering
Languages: SystemVerilog, VHDL, Python, C/C++, MATLAB, RISC-V assembly
Concepts: Comparator design, CDAC layout, AC, DC, and transient analysis, Monte Carlo & transient-noise analysis, offset/mismatch analysis, CMOS fundamentals, UVM, coverage-driven verification, test plan development